

3.2 Adaptive non-target support

The council should capture complementary predictions among alternatives that its members consider likely. We therefore define a shared non-target support at each reasoning position, adapting its size to the predictions of individual experts. Taking the union of their candidate sets preserves alternatives favored by any member and provides a common space in which to compare their predictions.

At position t , let $z_{m,t,j}^\downarrow$ and $v_{m,t,j}$ denote the j th-largest logit and its token identity among $\mathcal{V} \setminus \{y_t\}$ for expert m . We search only the top $K = \min(512, |\mathcal{V}| - 1)$ non-target candidates. Both rank and logit axes are normalized within this window, so the criterion identifies a local knee in the plausible head region rather than a global vocabulary knee. We use a locally normalized bounded Kneedle-style criterion (Satopää et al., 2011):

$$\begin{aligned} x_j &= \frac{j-1}{K-1}, \quad u_{m,t,j} = \frac{z_{m,t,j}^\downarrow - z_{m,t,K}^\downarrow}{z_{m,t,1}^\downarrow - z_{m,t,K}^\downarrow + \epsilon_{\text{sel}}}, \\ \hat{k}_{m,t} &= \arg \max_{1 \leq j \leq K} [(1 - x_j) - u_{m,t,j}], \\ k_{m,t} &= \min(K, \max(\hat{k}_{m,t}, 8)). \end{aligned} \quad (1)$$

Here $\epsilon_{\text{sel}} > 0$ ensures a well-defined logit normalization. The first maximizing rank is chosen when there is a tie. For $K = 1$, we set $\hat{k}_{m,t} = k_{m,t} = 1$ directly. The search window is the only upper bound; the lower bound of eight avoids impoverished supports whenever enough candidates are available. The criterion identifies the largest deviation of the locally normalized logit curve from the descending diagonal.

The council support is

$$\mathcal{C}_t = \bigcup_{m=1}^M \{v_{m,t,j} : 1 \leq j \leq k_{m,t}\}. \quad (2)$$

All experts are evaluated on these common token identities, and the support is held fixed when differentiating the diversity objective. Excluding the observed token y_t directs this objective toward alternative continuations of the reasoning trace.

3.3 Stage 1: Learning a diverse council

Stage 1 learns a council that combines fidelity to the observed traces with complementary predictions. We train M independently initialized LoRA adapters on the same fixed trajectories. Let $\theta_m = \{A_m^l, B_m^l\}_{l \in \mathcal{L}}$ denote the parameters of expert m . The training rule combines supervised learning, a determinantal objective in prediction space, and a kernel interaction in effective-weight space.

Trace supervision. Each expert learns from the full observed response through supervised fine-tuning. Let $\mathcal{T}_n^{\text{resp}}$ denote its response positions and $p_{m,n,t}$ the predictive distribution conditioned on the problem and observed prefix. The likelihood objective is

$$\mathcal{L}_{\text{SFT}}^{(m)} = -\frac{1}{N_{\text{resp}}} \sum_{n \in \mathcal{B}} \sum_{t \in \mathcal{T}_n^{\text{resp}}} \log p_{m,n,t}(y_{n,t}), \quad (3)$$

where $N_{\text{resp}} = \sum_{n \in \mathcal{B}} |\mathcal{T}_n^{\text{resp}}|$. This term anchors each expert to the supplied reasoning and answer, while the council interactions encourage complementary behavior among alternatives.

Diversity in prediction space. We measure predictive diversity on the non-target support from §3.2. For each reasoning position, define the normalized feature vector

$$v_{m,t}(v) = \frac{p_{m,t}(v)}{\sqrt{\sum_{u \in \mathcal{C}_t} p_{m,t}(u)^2}}, \quad v \in \mathcal{C}_t. \quad (4)$$

This representation depends on the relative probabilities of the alternatives, independently of their total mass. Let $\mathbf{V}_t$ stack these vectors as rows. For reasoning step s of example n , with content positions $\mathcal{T}_{n,s}^{\text{r}}$, define

$$G_{n,s} = \frac{1}{|\mathcal{T}_{n,s}^{\text{r}}|} \sum_{t \in \mathcal{T}_{n,s}^{\text{r}}} \mathbf{V}_t \mathbf{V}_t^\top. \quad (5)$$

This is the Gram matrix of the experts' concatenated, normalized step representations. We encourage these representations to span distinct directions using a

determinantal penalty (Kulesza and Taskar, 2012):

$$\begin{aligned}\ell_{n,s} &= -\frac{1}{M} \log \det(G_{n,s} + \epsilon_{n,s} \mathbf{I}), \\ \mathcal{L}_{\text{DPP}} &= \frac{1}{|\mathcal{B}_r|} \sum_{n \in \mathcal{B}_r} \frac{1}{|S_n^r|} \sum_{s \in S_n^r} \ell_{n,s}.\end{aligned}\quad (6)$$

Here $\epsilon_{n,s} > 0$ is a small diagonal regularizer, S_n^r contains the nonempty reasoning steps, and $\mathcal{B}_r$ contains the examples with such steps. The determinant acts on the step-averaged Gram matrix, encouraging complementary predictions across the reasoning step. Equal weighting across steps and examples limits the influence of differences in trace length. This objective applies only to reasoning content; SFT supervises the full response.

Diversity in effective-weight space. The LoRA factorization is nonunique: for any invertible R , $(B_m^l R)(R^{-1} A_m^l) = B_m^l A_m^l$. We therefore compare experts through their effective updates $\Delta W_m^l = (\alpha/r) B_m^l A_m^l$, using

$$d_{mq}^2 = \frac{1}{|\mathcal{L}|} \sum_{l \in \mathcal{L}} \frac{\|\Delta W_m^l - \Delta W_q^l\|_F^2}{d_{\text{out}}^l d_{\text{in}}^l}. \quad (7)$$

Normalizing by the matrix size gives each adapted projection equal weight. The resulting distance is invariant to refactorizations that preserve the effective update.

An RBF kernel $K_{mq} = \exp(-d_{mq}^2/h)$ expresses proximity between experts, where the positive bandwidth h follows a smoothed median of their pairwise distances. Inspired by kernel-based particle methods (Liu and Wang, 2016), we define a similarity potential and a repulsive direction:

$$\Phi = \frac{1}{\binom{M}{2}} \sum_{m < q} K_{mq}, \quad r_m = -\nabla_{\theta_m} \Phi. \quad (8)$$

With h held fixed, r_m is a local descent direction for pairwise similarity, providing a second diversity signal alongside the prediction-space objective.

Coupled council update. Let $g_m^{\text{SFT}} = \nabla_{\theta_m} \mathcal{L}_{\text{SFT}}^{(m)}$ and $g_m^{\text{DPP}} = \nabla_{\theta_m} \mathcal{L}_{\text{DPP}}$. The kernel coordinates

gradient exchange across experts:

$$\begin{aligned}\omega_{qm} &= \frac{K_{qm}}{\sum_{j=1}^M K_{jm}}, \\ a_m &= \sum_{q=1}^M \omega_{qm} (g_q^{\text{SFT}} + \lambda_{\text{div}} g_q^{\text{DPP}}).\end{aligned}\quad (9)$$

The weights form a convex combination for each receiving expert, favoring updates from members with similar effective weights. Gradients are combined in corresponding LoRA factor coordinates. We limit the repulsive direction relative to this aggregate by setting

$$\tilde{r}_m = \min \left\{ 1, \frac{\|a_m\|_2}{\|r_m\|_2 + \delta} \right\} r_m, \quad (10)$$

where $\delta > 0$ is a small stabilizer. Thus $\|\tilde{r}_m\|_2 \leq \|a_m\|_2$.

A scheduling coefficient $\gamma \in [0, 1]$ introduces the council interactions gradually after an initial period of independent SFT. The resulting pseudo-gradient is

$$\begin{aligned}g_m &= (1 - \gamma) g_m^{\text{SFT}} \\ &\quad + \gamma (a_m - \lambda_{\text{rep}} \tilde{r}_m).\end{aligned}\quad (11)$$

The coefficients λ_{div} and λ_{rep} control the strengths of predictive diversity and weight-space repulsion. Kernel weights and the repulsion rescaling are fixed when assembling this direction. Together, gradient exchange and repulsion balance shared trace supervision with complementary expert behavior.

References

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